EX10 - Rocket Equation with Functions - Defaults and Keywords

Beginner Step-by-Step Guide

This guide follows the real file `ex10_rocket_equation_function.py`. It explains the problem first and then the code from the perspective of someone starting Python.

1. What we want to solve

Encapsulate the rocket equation in a function and introduce a default g0 parameter.

Problem variables / elements

Name	Meaning
Isp	specific impulse
m0	initial mass
mf	final mass
g0	standard gravity

Equation or logic on paper

```
delta_v = Isp * g0 * ln(m0/mf)
```

Manual check

With Isp=250, m0=500000, mf=250000 and g0=9.80665, delta_v is about 1699.6 m/s.

2. What we already know and what is new

Already seen:

- def
- return
- None
- function calls

New in this exercise:

- default parameter
- positional arguments
- keyword arguments
- overriding a default
- test cases

3. New Python concepts

1. Default

Syntax: `g0 = 9.80665`

In the definition it means: if the caller does not supply g0, use 9.80665.

2. Positional

Syntax: `calculate_rocket_delta_v(Isp, m0, mf)`

Arguments are assigned to parameters by order.

3. Keyword

Syntax: `calculate_rocket_delta_v(Isp=500, m0=500000, mf=250000)`

Arguments are explicitly assigned by name.

4. Code step by step

Step 1. Define interface

```
def calculate_rocket_delta_v(Isp, m0, mf, g0=9.80665):
```

Isp, m0 and mf are required; g0 may be omitted.

Step 2. Validate

```
if Isp < 0:
    return None
elif m0 <= 0:
    return None
...
```

Each return stops that call if an invalid condition is met.

Step 3. Calculate

```
mass_ratio = m0 / mf
delta_v = Isp * g0 * np.log(mass_ratio)
return delta_v
```

This block is reached only if all previous validation conditions were False.

Step 4. Positional call

```
delta_v = calculate_rocket_delta_v(Isp, m0, mf)
```

g0 is omitted and therefore uses its default.

Step 5. Keyword call

```
calculate_rocket_delta_v(Isp=500, m0=500000, mf=250000)
```

Names make the argument-to-parameter mapping explicit.

Step 6. Test invalid inputs

```
calculate_rocket_delta_v(Isp, 250000, mf)
calculate_rocket_delta_v(Isp, m0, 0)
```

The same function can be used to see how different invalid cases are handled.

5. Execution flow

```
define function with default
 -> call
 -> validate
 -> compute
 -> return value or None
```

6. Expected result / check

The nominal call computes ~1699.6 m/s. Invalid calls print messages and return None.

7. Common beginner mistakes

- Placing a required parameter after a default one.
- Confusing defaults with keyword arguments.
- Not realizing return stops the function.
- Not checking None.

8. If you remember only five things

- A default makes a parameter optional at call time.
- Positional matches by order.
- Keyword matches by name.
- return ends the current call.
- A function can be reused for many cases.

9. Mini-practice

Change one input. Before running, write a comment predicting what should change and why. Then run the file and compare your prediction with the output.

Completion criterion

Move on when you can explain the problem, equation/logic, new Python concepts and execution flow without simply reading the code.